A Novel Moving Target Detection Method Based on RPCA for SAR Systems

Yifan Guo, Guisheng Liao, *Senior Member, IEEE*, Jun Li, *Member, IEEE*, and Xixi Chen

***Abstract*— Clutter background suppression and velocity estimation for moving targets are two critical problems in synthetic aperture radar–ground moving target indication (SAR-GMTI). A robust principal component analysis (RPCA) method is used to separate the sparse matrix of moving targets from the low-rank matrix of static backgrounds by using amplitude information in the image domain. However, the nonsparsity of the moving target echoes limits the performance of the RPCA in SAR-GMTI, and the velocity of the moving target cannot be estimated since the phase information is destroyed by the soft-thresholding operator in the RPCA process. To solve these problems, a novel moving target detection method based on RPCA (NRPCA) for SAR systems is proposed in this article. An atomic norm-based optimization program is first constructed to transform the data sparsity requirement into a moving target sparsity requirement. Although this optimization program is NP-hard, it is transformed to semidefinite programming by relaxation. Furthermore, accurate velocity estimation is performed using dual function theory and the alternating direction method of multipliers (ADMM) algorithm while the selection of the sparsity order k is avoided. Simulations and analyses based on experimental data illustrate the effectiveness of the proposed method.**

***Index Terms*— Ground moving target indication (GMTI), parameter estimation, robust principal component analysis (RPCA) method, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar–ground moving target indication (SAR-GMTI) systems are widely employed in traffic control, battlefield surveillance, and other applications. The velocity estimation of moving targets is a key issue in many synthetic aperture radar (SAR) applications [1], [2]. Most classic algorithms for velocity estimation are based on the difference of the azimuth (cross-range) phase among channels [3]–[5]. For instance, the radial velocity could be obtained from the interferometric phase between two images [4] or by the direction-of-arrival (DOA) estimation method [5]. However, it is difficult to directly detect the moving targets in SAR systems since the weak target is submerged in the strong clutter. Therefore, the clutter usually requires suppression before the velocity estimation of the moving target. The existing clutter suppression techniques are mainly divided into two types: "single-channel SAR" [6]–[8] and "two-channel or multichannel SAR" [9], [10]. The displaced phase center antenna (DPCA) and along-track interferometry (ATI) approaches are typical examples of two-channel SAR [11], [12]. In order to reduce the limitations of the platform parameters, multichannel DPCA [13] and ATI [14] have been proposed instead of the conventional DPCA and ATI methods. Beyond that, the research shows that many factors could affect the performance of the above methods. For example, a small moving target is difficult to be detected under the condition of strong stationary clutter, and the restriction of channel spacing could be not satisfied in practice. With the improvement in computing power, the space-time adaptive processing (STAP) method is suggested as a key enabling technology in [15] and [16] as an effective method to detect small, low-velocity targets when strong clutter and jamming interference exist. However, the traditional STAP requires a large amount of target-free homogeneous secondary data, which cannot always be obtained in practice. To overcome the requirement for a number of samples, a novel method combining compressive sensing (CS) algorithm and STAP is proposed [17]. Unfortunately, the user setting parameters required in the CS algorithm are not easy to obtain. In addition, the computational complexity of the CS method increases considerably with the increasing measurement vector dimensions. In order to reduce the computational burden via the matrix processing of data, a low-rank matrix recovery (LRMR) theory for 2-D matrix data is proposed in [18] and [19]. A robust principal component analysis (RPCA) [20] is one of the sparse optimization models for LRMR, which has been applied widely to the image and video processing [21]. The purpose of RPCA is to divide an approximate low-rank matrix into a low-rank component, a strong sparse component, and a weak noise background component [22]. However, only the amplitude information is used to get the results in the field of image and video processing. Unfortunately, the important information of the moving targets lies in the phase for SAR systems. In [23] and [24], the echo data are directly processed to divide ground moving targets by the RPCA after the pulse compression. However, the background echo does not always satisfy the low-rank characteristic in the strong noise power

Manuscript received May 21, 2019; revised December 1, 2019 and January 10, 2020; accepted February 23, 2020. Date of publication March 16, 2020; date of current version August 28, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61431016 and Grant 61621005. *(Corresponding author: Yifan Guo.)*

Yifan Guo, Guisheng Liao, and Jun Li are with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China (e-mail: yifanguo@163.com).

Xixi Chen is with the College of Electronic Science, National University of Defense Technology, Changsha 410073, China.

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2978496

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

such that the target energy leaks into the clutter background. In [25], an approximate low-rank matrix is structured by the subapertures in the azimuth direction, where the structured data have a superior low-rank characteristic. Subsequently, the moving target imaging is separated from the background. As only the amplitude information is used, the result is not satisfactory. In [26]–[28], the phase information is translated into the envelope. Then, the moving targets are indicated by the RPCA method. However, these methods need complex preprocessing. Due to the phase difference among the along-track interferometric channels caused by the radial velocity of the moving target, an RPCA-based moving target detection method is introduced in [29]–[31] for a multichannel SAR system. Then, the moving target can be effectively separated from the clutter in the low-rank background. However, the SAR-GMTI based on the RPCA has two problems. First, the sparsity assumption of the moving target echo in the range-Doppler domain is not necessarily satisfied for multiple targets in practice. Second, the phase information of the echo data is destroyed by the soft-thresholding operator in the processing of the RPCA, which causes the target velocity to be inestimable. In [32], certain initial values of the low-rank and sparse matrices are specified for the RPCA method. However, the above two problems have not been resolved yet. In [33], an RPCA method based on the constructed 4-D dictionary matrix is performed. The velocity of the moving targets is estimated, and the sparsity requirement of the sparse matrices is reduced by this method. However, the estimated velocity is affected by the accuracy of the dictionary matrix, and a high-dimension dictionary matrix will lead to a high computational complexity.

To handle the above two problems and improve the accuracy of the estimated velocity, we propose a novel moving target detection method based on RPCA (NRPCA) for a multichannel SAR model, which is an improved method for RPCA. For the first problem, an atomic norm-based optimization program is constructed after the data sparsity requirement is transformed to a moving target sparsity requirement. Unfortunately, this optimization program is NP-hard. In order to solve this optimization problem, we transform this NP-hard problem to a semidefinite programming problem by relaxation. Theoretically, the semidefinite programming problem can be solved by using the interior-point method (IPM) [34]. However, it is while known that the IPM requires a large memory space, and the computation burden is too high to solve large-scale matrix problems such as the SAR-GMTI. Although the soft-thresholding operator can be used to effectively reduce the computation burden [35], the phase information is destroyed by the soft-thresholding operator in the processing of the RPCA. Therefore, we propose a method based on the alternating direction method of multipliers (ADMM) algorithm to offset the shortcomings of the IPM and soft-thresholding operator. For the second problem, the velocity of a moving target can be obtained by the solved dual problem of the primary problem. In addition, compared with the method of directly solving l_0 norm minimum problem, we utilize solving its dual problem to avoid the selection of the sparsity order k and the grid point error. In the proposed method, the moving target can be separated from the background clutter. Meanwhile, the velocity of the moving target is accurately estimated by using the phase information of the echo data under the condition of a sampling regime. The simulation results confirm the effectiveness of the proposed method.

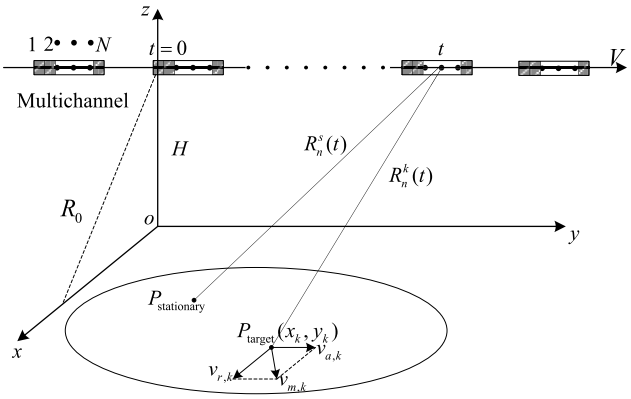


Fig. 1. Multichannel SAR system.

The remainder of this article is organized as follows. In Section II, we introduce the SAR imaging geometry and the data model for this SAR-GMTI problem, and the optimization model of the NRPCA is proposed in Section III. In Section IV, the optimization problem is solved. A simulation study is presented in Section V. In Section VI, we present the conclusions.

Notations: We denote vectors and matrices by boldface lower- and uppercase letters, respectively. $(\cdot)^T$ represents the transpose operation, and rank$(\cdot)$ represents the rank. $\|\cdot\|_0$, $\|\cdot\|_{\mathcal{A}}$, $\|\cdot\|_*$, $\|\cdot\|_2$, $\|\cdot\|_\infty$, and $\|\cdot\|_F$ describe the l_0 norm, atomic norm, nuclear norm, l_2 norm, l_∞ norm, and Frobenius norm, respectively.

II. Signal Model

We consider a multichannel SAR system with the geometry shown in Fig. 1.

Assuming that there are N channels. The first channel is the transmitted channel, and all others are the received channels. The velocity and height of the radar platform are denoted by $V$and H, respectively. λ represents the radar wavelength. f_c denotes the carrier frequency, whereas d is the interelement spacing. $R_k = (H^2 + x_k^2)^{1/2}$ is the instant range when the radar boresight is directed to the kth target. $R_n^s(t)$ is the distance between the nth channel and the clutter point $P_{\text{stationary}}$ at a time t. $R_n^k(t)$ represents the distance between the nth channel and the moving target $P_{\text{target}}(x_k, y_k)$ at a time t. We take the location of the aircraft $t = 0$ on the ground, O, as the origin of the coordinate system. The y-axis points in the same direction as the velocity vector of the aircraft, and the coordinates of the receiver are $(0, 0, H)$. The xoy plane is on the ground plane, and the z-axis points vertically upward. u and t denote the fast time in the range dimension and the slow time in the azimuth dimension, respectively. $v_{m,k}$ is the velocity of the moving target, whereas $v_{r,k}$ and $v_{a,k}$ denote the radial velocity and azimuth velocity of the kth moving target, respectively. Suppose $V \approx V - v_{a,k}$; then, the instantaneous slant range

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

$R_n^k(t)$ of the kth target can be expressed as

$$R_n^k(t) \approx R_k + v_{r,k}(t - y_k/V) + \frac{(Vt - y_k + d(n-1))^2}{2R_k}. \quad (1)$$

After range pulse compression, the echo of the nth channel and the kth target can be formulated as

$$s_n^k(u,t) = \sigma_k p_k(u) w_a(t) \exp\left(-j\frac{2\pi}{\lambda}\left(R_1^k(t) + R_n^k(t)\right)\right) \quad (2)$$

where σ_k is the reflection coefficient for the kth target, $p_k(u)$ denotes the range envelope after the range pulse compression, and $w_a(t)$ is the azimuth envelope before the azimuth compression. (x_k, y_k) is the position of the kth target when the radar boresight is directed to the kth target. Hence, the received echoes in a channel n are a linear combination of each scattering point in this scene, namely

$$\begin{aligned}\mathbf{S}_n &= \sum_k s_n^k(u,t) = \sum_k \sigma_k p_k(u) w_a(t) \\ &\quad \times \exp\left(-j\frac{2\pi}{\lambda}\left[2R_k + 2v_{r,k}(t - y_k/V) + \frac{(Vt - y_k)^2}{2R_k}\right.\right. \\ &\quad \left.\left. + \frac{(Vt - y_k + d(n-1))^2}{2R_k}\right]\right). \quad (3)\end{aligned}$$

Substituting the stationary phase into the Fourier transform (FT) integral, (3) in the 2-D frequency domain can be gotten as follows:

$$\begin{aligned}\mathbf{S}_n &= \sum_k s_n^k(f_r, f_a) = \sum_k \sigma_k p_k(f_r) w_a(f_a) \\ &\quad \times \exp\left(j\frac{\pi f_a^2 R_k c}{2V^2(f_c + f_r)}\right) \\ &\quad \times \exp\left(-j\frac{2\pi(f_c + f_r)}{c}\left(2R_k + \frac{v_{r,k}^2 R_k}{V^2}\right.\right. \\ &\quad \left.\left. + \frac{v_{r,k}(n-1)d}{V} - \frac{(d(n-1))^2}{4R_k}\right)\right) \\ &\quad \times \exp\left(j\frac{2\pi f_a v_{r,k} R_k}{V^2} + j\frac{\pi f_a (n-1)d}{V} - j\frac{2\pi f_a y_k}{V}\right) \quad (4)\end{aligned}$$

where f_r and f_a denote the range frequency and Doppler frequency, respectively. All channels have similar received data except for the phase part caused by the channel spacing d. The different channel data are compensated by the function

$$H(f_a) = \exp\left(-j\pi\frac{(n-1)d}{V}f_a + j\frac{\pi(f_c + f_r)((n-1)d)^2}{2cR_o}\right) \quad (5)$$

where R_0 is the shortest distance between the middle range gate and the radar moving track. After the compensation in the 2-D frequency domain, (4) can be expressed in the 2-D time domain as

$$\begin{aligned}&\mathbf{S}_n(t,u) \\ &= \sum_k \sigma_k p(u - u_k) w_a(t) \\ &\quad \times \exp\left(-j\frac{4\pi}{\lambda}\left(R_k + v_{r,k}(t - y_k/V) + \frac{(Vt - y_k)^2}{2R_k}\right)\right) \\ &\quad \times \exp\left(j\frac{2\pi(n-1)dv_{r,k}}{\lambda V}\right) \quad (6)\end{aligned}$$

where

$$u_k = \frac{2}{c}\left(R_k + v_{r,k}(t - y_k/V) + \frac{(Vt - y_k)^2}{2R_k}\right) - \frac{(n-1)dv_{r,k}}{cV}$$

and

$$\frac{\pi(f_c + f_r)((n-1)d)^2}{2cR_k} - \frac{\pi(f_c + f_r)((n-1)d)^2}{2cR_o} \approx 0.$$

By comparing the data in each channel after the compensation, there is only a phase difference consisting of $\exp(j(2\pi(n-1)dv_{r,k}/\lambda V))$ for the moving target. Moreover, the phase information is contained in the envelope $p(u - u_k)$. However, there is no phase difference or delay information for the clutter background in different channels. Thus, the moving target velocity can be estimated by using this phase difference after the clutter suppression. Here, $f_v = dv_r/\lambda V$ is defined as the velocity frequency.

Vectorizing the received signals and stacking all received channel signals as a matrix, we obtain

$$\mathbf{Z} = [\mathrm{vec}(\mathbf{S}_1(t,u))\ \mathrm{vec}(\mathbf{S}_2(t,u))\ \ldots\ \mathrm{vec}(\mathbf{S}_n(t,u))]^T \quad (7)$$

where $\mathrm{vec}(\cdot)$ denotes the vectorization operator. In this case, the received signal can be reformulated as

$$\mathbf{Z} = \mathbf{A} + \mathbf{D} = \mathbf{A} + \mathbf{LMR} \quad (8)$$

where $\mathbf{A} = \mathbf{Z}|_{v_{r,k}=0}$ ($\mathbf{A} \in \mathbb{R}^{N \times PQ}$) with $\mathbf{A}$ representing the received data of the static background clutter. P is the fast-time sample number, and Q is the slow-time sample number. f_s is the sampling frequency, and PRF is the pulse repetition frequency. $\mathbf{D} \in \mathbb{R}^{N \times PQ}$ is the received data matrix of the moving targets

$$\begin{aligned}\mathbf{D} = \mathbf{LMR} &= \sum_k \sigma_k p(u) w_a(t) \\ &\quad \times \exp\left(-j\frac{4\pi}{\lambda}\left(R_k + v_{r,k}t + \frac{(Vt - y_k)^2}{2R_k}\right)\right) \\ &\quad \times \exp\left(-j\frac{2\pi(n-1)dv_{r,k}}{\lambda V}\right) \quad (9)\end{aligned}$$

where

$$\mathbf{M} = \mathrm{diag}[\sigma_1\ \sigma_2\ \ldots\ \sigma_K] \quad (10)$$

$$\mathbf{L} = [\mathbf{l}(v_1)\ \mathbf{l}(v_2)\ \ldots\ \mathbf{l}(v_K)] \quad (11)$$

$$\mathbf{R} = [\mathbf{r}(R_1)\ \ \mathbf{r}(R_2)\ \ldots\ \mathbf{r}(R_K)]^T\ \mathbf{R} \in \mathbb{R}^{K \times PQ}. \quad (12)$$

where $\mathbf{M}$ is a square matrix, K is the maximum number of moving targets, $\mathbf{L}$ denotes the left matrix formed by stacking the vector of the kth moving target phase caused by interelement spacing

$$\mathbf{l}(v_{r,k}) = \left[\exp\left(j\frac{2\pi dv_{r,k}}{\lambda V}\right)\ \ldots\ \exp\left(j\frac{2\pi(n-1)dv_{r,k}}{\lambda V}\right)\right]^T$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

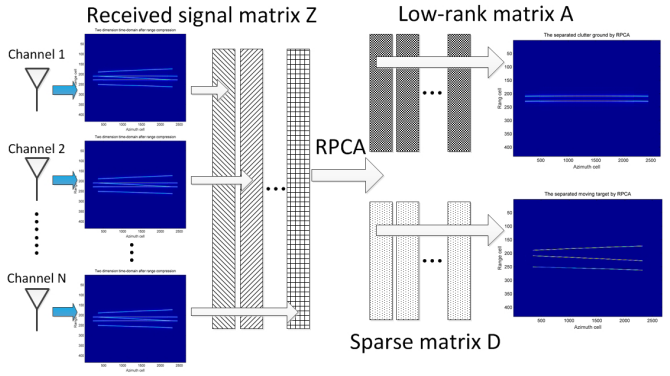


Fig. 2. Flowchart of the RPCA method.

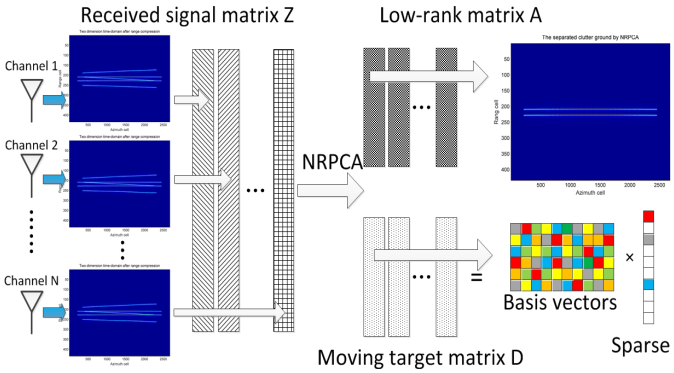


Fig. 3. Flowchart of the proposed method.

and $\mathbf{R}$ denotes the right matrix constructed by stacking the vector of the residual phase for the kth moving target

$$\begin{aligned}&\mathbf{r}(R_k)\\&= \Bigg[p(u-u_k)w_a(t)\\&\quad\times\exp\left(-j\frac{4\pi}{\lambda}\left(R_k+v_{r,k}t+\frac{(Vt-y_k)^2}{2R_k}\right)\right)|_{u=0,t=0}\\&\quad\ldots\ p(u-u_k)w_a(t)\\&\quad\times\exp\left(-j\frac{4\pi}{\lambda}\left(R_k+v_{r,k}t+\frac{(Vt-y_k)^2}{2R_k}\right)\right)|_{u=P/f_s,t=QT}\Bigg]^T.\end{aligned}$$

III. Improvement and Analysis of the Algorithm

In this section, we first analyze the shortcomings of the RPCA method. Then, the NRPCA method for overcoming these shortcomings is proposed, and the optimization model of the NRPCA is illustrated in detail.

The essence of the RPCA is a procedure of decomposing a known matrix into a low-rank matrix and a sparse matrix, according to [20]. After channel image registration, pulse compression, and phase compensation, all channels have similar received data, except that they have amplitude-phase differences caused by the radial velocity of the moving target. Therefore, the background clutter can ideally be stacked as a rank-one matrix $\mathbf{A}$. If the sampled data of the target echo are sparse relative to the total sampled data, the target echo data will still have a sparsity in the range-Doppler domain where the moving target echo data are expressed as the matrix $\mathbf{D}$. Based on these features, the RPCA method can effectively separate the moving target from the background clutter when it is applied to the SAR-GMTI, as described in [31]. Unfortunately, the phase information will be degraded in [31]. Hence, the RPCA method cannot be utilized to effectively estimate the motion parameters. A flowchart of this process is shown in Fig. 2.

In Fig. 2, the problem of separating a moving target from the clutter background can be summarized as an optimization problem

$$\begin{aligned}&\min_{\mathbf{A},\mathbf{D}}\ \operatorname{rank}(\mathbf{A}),\quad \|\mathbf{D}\|_0\\&\text{s.t. }\mathbf{Z}=\mathbf{A}+\mathbf{D}.\end{aligned}\tag{13}$$

It can be observed from (13) that $\mathbf{D}$ should be sparse to resolve this optimization problem. However, the echo data could not practically be sparse in the full-dimensional space even though the target is sparse relative to all scatter points with an increased number of targets. To handle this problem, we utilize matrix $\mathbf{M}$ sparse to improve the traditional method. The optimization problem can be modified as

$$\begin{aligned}&\min_{\mathbf{A},\mathbf{D}}\ \operatorname{rank}(\mathbf{A}),\quad \|\mathbf{M}\|_0\\&\text{s.t. }\mathbf{Z}=\mathbf{A}+\mathbf{LMR}.\end{aligned}\tag{14}$$

In (14), the velocity information is contained in the basis matrix $\mathbf{L}$. However, the accuracy of the estimated velocity is limited by the accuracy of the divided velocity grid set in (14). To avoid restrictions caused by the accuracy of the discretization grids, we propose the NRPCA algorithm

$$\begin{aligned}&\min_{\mathbf{A},\mathbf{D}}\ \operatorname{rank}(\mathbf{A}),\quad \|\mathbf{D}\|_{\mathcal{A}}\\&\text{s.t. }\mathbf{Z}=\mathbf{A}+\mathbf{D}\\&\quad\mathbf{D}=\mathbf{LMR}\end{aligned}\tag{15}$$

where $\|\mathbf{D}\|_{\mathcal{A}}=\inf_{f_{v,k}\in[0,1]}\{\sum_k\sigma_k:\mathbf{D}=\sum_k\sigma_k\mathbf{l}^H(f_{v,k})\mathbf{r}(\phi_k)\}$ and $\hat{\mathbf{D}}$ is the solution for the optimization problem $\min_{\mathbf{D}}\ \|\mathbf{D}\|_{\mathcal{A}}$ in the continuous domain. Aforementioned, $\min_{\mathbf{D}}\ \|\mathbf{D}\|_0$ means the sparsity requirement restriction of the echo data $\mathbf{D}$. However, by substituting $\min_{\mathbf{D}}\ \|\mathbf{D}\|_{\mathcal{A}}$ for $\min_{\mathbf{D}}\ \|\mathbf{D}\|_0$, it transforms the sparsity requirement of the echo data into the moving target sparsity requirement. Compared with $\min_{\mathbf{M}}\ \|\mathbf{M}\|_0$, $\min_{\mathbf{D}}\ \|\mathbf{D}\|_{\mathcal{A}}$ is solved in the continuous domain of independent variable $f_{v,k}$ such that it has no grid point error. Therefore, the moving target matrix $\mathbf{D}$ is accurately separated from the echo data $\mathbf{Z}$. However, in this case, the sampling regime must be satisfied for the velocity frequency f_v of different targets. This sampling regime is formally described by Candès and Fernandez-Granda [36]. Based on the idea of the off-grid CS theory, a flowchart of the NRPCA method is shown in Fig. 3.

IV. Solved Optimization Problem

As optimization problem (15) is an NP-hard problem, we will show how to solve the optimization problem (15) in this section. A positive weight $\lambda>0$ is introduced, and (15)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

can be reformulated as

$$\min_{\mathbf{A},\mathbf{D}} \operatorname{rank}(\mathbf{A}) + \lambda \cdot \|\mathbf{D}\|_{\mathcal{A}} \quad \text{s.t. } \mathbf{Z} = \mathbf{A} + \mathbf{D}. \tag{16}$$

Since the optimization of $\min_{\mathbf{A}} \operatorname{rank}(\mathbf{A})$ is an NP-hard problem, we should relax the optimization problem as $\min_{\mathbf{A}} \|\mathbf{A}\|_*$. Thus, (16) can be rewritten into

$$\min_{\mathbf{A},\mathbf{D}} \|\mathbf{A}\|_* + \lambda \cdot \|\mathbf{D}\|_{\mathcal{A}} \quad \text{s.t. } \mathbf{Z} = \mathbf{A} + \mathbf{D}. \tag{17}$$

The generalized Lagrange function of the above optimization problem with restrictions is given by

$$L(\mathbf{A},\mathbf{D},\mathbf{Y},\mu,\lambda) = \|\mathbf{A}\|_* + \lambda \cdot \|\mathbf{D}\|_A + \operatorname{tr}\left(\mathbf{Y}^H(\mathbf{Z}-\mathbf{A}-\mathbf{D})\right) + \frac{\mu}{2} \cdot \|\mathbf{Z}-(\mathbf{A}+\mathbf{D})\|_F^2 \tag{18}$$

where μ is a penalty factor, $\mathbf{Y}$ represents the Lagrange multiplier, and $\operatorname{tr}(\cdot)$ denotes the trace operator. Afterward, we use the ADMM algorithm to solve (18). First, we set the initialization parameters. μ and λ should be greater than zero. In general, their values are constants obtained from empirical values. When penalty parameters μ^k and λ^k varying with the number of iterations are selected, the optimization problem has a faster convergence speed. However, their performance is similar. This approach is analyzed in [37] and [38]. In this article, $\mu = 0.5, \lambda = 0.4, \mathbf{A}_0 = \mathbf{0}, \mathbf{D}_0 = \mathbf{0}$, and $\mathbf{Y}_0 = (\mathbf{Z}/\max(\|\mathbf{Z}\|_2, \lambda^{-1}\|\mathbf{Z}\|_\infty))$. Then, the Lagrange function of $\mathbf{A}$ is expressed as follows:

$$\begin{aligned}\min_{\mathbf{A}_{k+1}} \; & L(\mathbf{A},\mathbf{D}_k,\mathbf{Y}_k,\mu,\lambda) \\ &= \|\mathbf{A}\|_* + \operatorname{tr}\left(\mathbf{Y}_k^H(\mathbf{Z}-\mathbf{A}-\mathbf{D}_k)\right) + \frac{\mu}{2} \cdot \|\mathbf{Z}-(\mathbf{A}+\mathbf{D}_k)\|_F^2 \\ &= \|\mathbf{A}\|_* + \frac{\mu}{2} \cdot \|\mathbf{A}-(\mathbf{Z}-\mathbf{D}_k+\mathbf{Y}_k/\mu)\|_F^2. \end{aligned} \tag{19}$$

The method for solving optimization problem (19) has been proven in [35] and is expressed as follows:

$$\mathbf{U}S_\varepsilon[\mathbf{S}]\mathbf{V}^T = D_\varepsilon(\mathbf{Q}) = \arg\min_{\mathbf{X}} \varepsilon\|\mathbf{X}\|_* + \frac{1}{2}\|\mathbf{X}-\mathbf{Q}\|_F^2 \tag{20}$$

where $\mathbf{USV}^{\mathrm{H}}$ is the singular value decomposition (SVD) form of $\mathbf{Q}$. The soft-thresholding operator needs to satisfy [35]

$$S_\varepsilon[x] = \begin{cases} x-\varepsilon, & \text{when } x > \varepsilon \\ x+\varepsilon, & \text{when } x < -\varepsilon \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

In this case

$$\hat{\mathbf{A}}_{k+1} = D_{1/\mu}(\mathbf{Z}-\mathbf{D}_k+\mathbf{Y}_k/\mu). \tag{22}$$

The Lagrange function of $\mathbf{D}$ is expressed as follows:

$$\begin{aligned}\min_{\mathbf{D}_{k+1}} \; & L(\mathbf{A}_{k+1},\mathbf{D},\mathbf{Y}_k,\mu,\lambda) \\ &= \lambda \cdot \|\mathbf{D}\|_{\mathcal{A}} + \operatorname{tr}\left(\mathbf{Y}_k^H(\mathbf{Z}-\mathbf{A}_{k+1}-\mathbf{D})\right) + \frac{\mu}{2} \cdot \|\mathbf{Z}-(\mathbf{A}_{k+1}+\mathbf{D})\|_F^2. \end{aligned} \tag{23}$$

Obviously, the following optimization is equivalent to (23):

$$\min_{\mathbf{D}_{k+1}} \; L(\mathbf{A}_{k+1},\mathbf{D},\mathbf{Y}_k,\mu,\lambda) = \lambda \cdot \|\mathbf{D}\|_{\mathcal{A}} + \frac{\mu}{2} \cdot \|\mathbf{A}_{k+1}-(\mathbf{Z}-\mathbf{D}+\mathbf{Y}_k/\mu)\|_F^2. \tag{24}$$

The phase information of matrix $\mathbf{A}$ is destroyed by the soft-thresholding operator in (20). Hence, substituting $\hat{\mathbf{A}}_{k+1}$ into (24), the phase information of the solved matrix $\mathbf{D}$ is destroyed too. Interestingly, we find from (6) that the phase information $\exp(j2\pi f_c u_k)$ is contained in the envelope $p(u-u_k)$. Thus, the phase of matrix $\hat{\mathbf{A}}_{k+1}$ can be easily recovered by the envelope of matrix $\hat{\mathbf{A}}_{k+1}$. By substituting the undestroyed $\hat{\mathbf{A}}_{k+1}$ into (24), the solved $\hat{\mathbf{D}}_{k+1}$ includes the phase information. In addition, we can observe that optimization problem (24) is an NP-hard problem. Therefore, according to the idea of off-grid CS theory [39], one can equivalently rewrite (24) as the following semidefinite program:

$$\begin{aligned}\min_{\mathbf{W},\mathbf{u},\mathbf{D}_{k+1}} \; & \operatorname{Tr}(\mathbf{W}) + \operatorname{Tr}(\mathbf{T}(\mathbf{u})) + \frac{\mu}{2} \cdot \|\mathbf{A}_{k+1}-(\mathbf{Z}-\mathbf{D}+\mathbf{Y}_k/\mu)\|_F^2 \\ \text{s.t. } & \begin{bmatrix} \mathbf{W} & \mathbf{D}^H \\ \mathbf{D} & \mathbf{T}(\mathbf{u}) \end{bmatrix} \succ 0 \end{aligned} \tag{25}$$

where $\hat{\mathbf{D}}_{k+1}$ is the solution to problem (25) and $\mathbf{T}(\mathbf{u})$ is the Toeplitz matrix whose first column is equal to $\mathbf{u}$. If (25) is directly solved by CVX [40], the solved velocity frequencies will still contain grid errors. Fortunately, one can recover the set of velocity frequencies from the solution of the dual problem of (24) to remove grid errors. The Lagrange dual function of (24) is

$$\begin{aligned} & g\left(\mathbf{A}_{k+1},\mathbf{Y},\mathbf{Y}',\mu,\lambda\right) \\ &= \inf_{\mathbf{D},\mathbf{U}} L'(\mathbf{A}_{k+1},\mathbf{D},\mathbf{U},\mathbf{Y}_k,\mathbf{Y}',\mu,\lambda) \\ &= \inf_{\mathbf{D},\mathbf{U}} \lambda \cdot \|\mathbf{U}\|_{\mathcal{A}} + \mathbf{Y}'^H(\mathbf{D}-\mathbf{U}) + \frac{\mu}{2} \cdot \|\mathbf{A}_{k+1}-(\mathbf{Z}-\mathbf{D}+\mathbf{Y}_k/\mu)\|_F^2 \\ &= \inf_{\mathbf{U}} \lambda \cdot \|\mathbf{U}\|_{\mathcal{A}} - \mathbf{Y}'^H\mathbf{U} + \inf_{\mathbf{D}} \mathbf{Y}'^H\mathbf{D} + \frac{\mu}{2} \cdot \|\mathbf{A}_{k+1}-(\mathbf{Z}-\mathbf{D}+\mathbf{Y}_k/\mu)\|_F^2 \\ &= \begin{cases} 0, & \|\mathbf{Y}'\|_* \le \lambda \\ -\infty, & \text{otherwise} \end{cases} \\ &\quad + \begin{cases} \mathbf{Y}'^H(\mathbf{Z}-\mathbf{A}_{k+1}+(\mathbf{Y}_k-\mathbf{Y}')/\mu) + \frac{\mu}{2} \cdot \|\mathbf{Y}'/\mu\|_F^2, \\ \qquad \mu \cdot (\mathbf{A}_{k+1}+\mathbf{D}-\mathbf{Z}) + \mathbf{Y}' - \mathbf{Y}_k = 0 \\ -\infty, \quad \text{otherwise.} \end{cases} \end{aligned} \tag{26}$$

Since (26) has now been transformed to a semidefinite problem, we observe that the optimization problem in (26) has a finite maximum if and only if

$$\|\mathbf{Y}'\|_* \le \lambda \qquad \mu \cdot (\mathbf{A}_{k+1}+\mathbf{D}-\mathbf{Z}) + \mathbf{Y}' - \mathbf{Y}_k = 0. \tag{27}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 NRPCA Algorithm for Solving (17)

Input: Multichannel echo data matrix $\mathbf{Z}$
Initialize: $\mu > 0$, $\lambda > 0$, $\mathbf{Y}_0 = \frac{\mathbf{Z}}{\max(\|\mathbf{Z}\|_2, \lambda^{-1}\|\mathbf{Z}\|_\infty)}$, $\mathbf{D}_0 = \mathbf{0}$
while not convergence **do**
1) $\hat{\mathbf{A}}_{k+1}$ of the damaged phase information is estimated by (19);
2) Re-estimate $\hat{\mathbf{Y}}'$ by (28);
3) Re-estimate $\hat{\mathbf{l}}_{k+1}$ by (30);
4) Reconstruct $\hat{\mathbf{D}}_{k+1}$ by (9)
5) Update the Lagrange multiplier $\mathbf{Y}_{k+1}$ by (31)
end while
Re-estimate $\hat{\mathbf{A}}$ of the contained phase information by $\hat{\mathbf{A}} = \mathbf{Z} - \hat{\mathbf{D}}$
Output: Estimated clutter echo data matrix $\hat{\mathbf{A}}$, estimated moving target velocity matrix $\hat{\mathbf{l}}$

Then, the dual problem of (24) is acquired as

$$\begin{aligned} &\max_{\mathbf{Y}'} \ \mathbf{Y}'^H(\mathbf{Z} - \mathbf{A}_{k+1} + (\mathbf{Y}_k - \mathbf{Y}')/\mu) + \frac{\mu}{2} \cdot \|\mathbf{Y}'/\mu\|_F^2 \\ &\text{s.t. } \|\mathbf{Y}'\|_* \le \lambda \\ &\quad \mu \cdot (\mathbf{A}_{k+1} + \mathbf{D} - \mathbf{Z}) + \mathbf{Y}' - \mathbf{Y}_k = 0. \end{aligned} \tag{28}$$

Here, the optimization problem (28) can be solved by using SDP

$$\hat{\mathbf{Y}}' = \mathbf{Y}_k - \mu \cdot (\mathbf{A}_{k+1} + \hat{\mathbf{D}}_{k+1} - \mathbf{Z}) \tag{29}$$

where $\hat{\mathbf{Y}}'$ is the solution to problem (28). Subsequently, (29) can be used to obtain the needed support set $\mathbf{S}$ of the moving target velocity via the following proposition, which is the solution to problem (24) in the continuous domain.

Proposition 1: Suppose that $\hat{\mathbf{Y}}'$ is the solution to problem (28). If each element in $\boldsymbol{\mathcal{A}}$ and a subset $\boldsymbol{\mathcal{S}} \subset \boldsymbol{\mathcal{A}}$ satisfy

$$\begin{cases} \langle \hat{\mathbf{Y}}', \mathbf{l} \rangle = \mathbf{r}_k, & \forall \mathbf{l} \in \mathbf{S} \\ \|\hat{\mathbf{Y}}', \mathbf{l}\|_2 < \|\mathbf{r}_k\|_2, & \forall \mathbf{l} \notin \mathbf{S} \end{cases} \tag{30}$$

then $\mathbf{S}$ is the support set of the solution to problem (24).

Proof: Please see Appendix A.

To update the Lagrange multiplier

$$\mathbf{Y}_{k+1} = \mathbf{Y}_k + \mu \cdot (\mathbf{Z} - \mathbf{D}_{k+1} - \mathbf{A}_{k+1}). \tag{31}$$

The solution $\mathbf{D}$ of the proposed optimization problem is convergent using the proposed method.

Proof: Please see Appendix B.

Since the phase information of the optimum solution in (19) is damaged, we obtain the background clutter with phase information according to (8), in which the moving target data $\hat{\mathbf{D}}$ is estimated from (9) and (30). Then, the background clutter with the phase information is given by $\hat{\mathbf{A}} = \mathbf{Z} - \hat{\mathbf{D}}$.

We summarize the proposed NRPCA algorithm in **Algorithm** 1 as follows.

V. Numerical Experiments

In this section, the proposed NRPCA, RPCA [31], ATI, and CS methods are compared and analyzed in terms of different aspects.

Through the analysis of the NRPCA and RPCA methods, there are two important factors affecting the detection results of the above two methods. The first factor is the sparsity. In [31], the optimization problem of a separated moving target is

$$\begin{aligned} &\min \|\mathbf{Z} - \mathbf{A} - \mathbf{D}\|_F^2 \\ &\text{s.t. } \text{rank}(\mathbf{A}) \le r \\ &\quad \text{card}(\mathbf{D}) \le k. \end{aligned} \tag{32}$$

In (32), the sparsity order k is defined as the number of nonzero elements in the sparse matrix $\mathbf{D}$ [31]. Specifically, in the practical application of moving target detection, the parameter k indicates the number of total resolution units occupied by the moving targets of interest, which usually cannot be known in advance. However, the selection of the sparsity order k exhibits a direct effect on the performance. When parameter k is set to a small value, a fewer points of the moving target could not be detected by the RPCA method. This will cause some information loss to the target. On the contrary, if the selected value of parameter k is larger, more points will be misidentified as target points such that the false-alarm rate will increase with parameter k. In the proposed method, the sparsity requirement of the moving target echo data will be replaced by the sparsity requirement of the moving target. In this case, the moving target will be extracted, regardless of whether the echo data of the moving target are sparse. Furthermore, by proposition 1, it can be obtained that only the moving target velocity satisfying (30) can be extracted to the support set $\mathbf{S}$ in the field of definition. Therefore, the resolution unit occupied by the targets does not need to be known beforehand, namely, the selection of the sparsity order k is avoided in the proposed method.

The second factor is the interchannel correlation and sparsity among channels. The interchannel correlation determines whether the background clutter matrix satisfies low-rank characteristics. Ignoring the noise, the rank of the clutter background matrix is 1 strictly under the condition of no interchannel errors, so the clutter background matrix has a stricter low-rank characteristic with a better interchannel correlation. Furthermore, strong noise, namely, when the signal-to-noise ratio (SNR) is small, will affect the sparsity among the channels such that high-magnitude noise will be mistaken as the moving target in the process of extracting sparsity matrix $\mathbf{D}$ by the RPCA method. However, the requirement of the echo data matrix sparse is transformed into that of the moving target sparse by the proposed method such that the NRPCA still works when the SNR is low. It is noteworthy that both the NRPCA and RPCA are based on the interchannel correlation. Hence, the signal-to-clutter ratio (SCR) has a little impact on the performance.

The effectiveness of the NRPCA method is verified by the point target simulation below. The parameters for the SAR operation used in this article are shown in Table I.

It is assumed that there are four static clutter points and three moving targets in a scene of interest, which are represented by squares and asterisks, respectively. The positions of these points are shown in Fig. 4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETERS FOR THE SAR OPERATION

Parameter		Value
Number of elements	N	10
Radar platform velocity	V	150 m/s
Height of the radar platform	H	6 km
Carrier wavelength	λ	0.03 m
Interelement spacing	d	0.3 m
Bandwidth	B	50 MHz
Sampling frequency	f_s	80 MHz
Pulse repetition frequency	PRF	1000 Hz

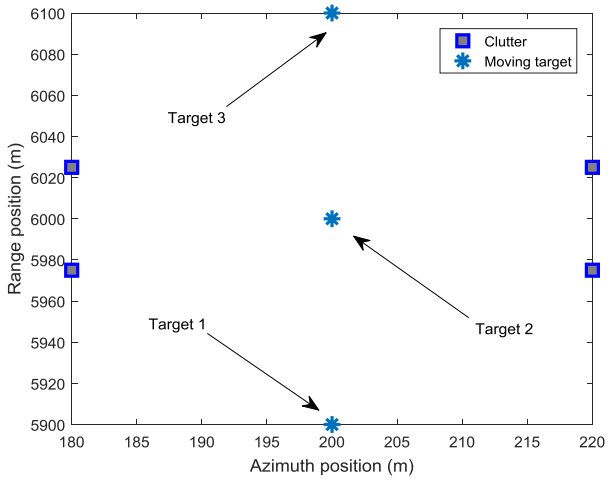


Fig. 4. Positions of clutter points and moving targets.

The radial velocities of the three moving targets are −15, 17.320, and 10.833 m/s, respectively. When ignoring interchannel error, SCR = 0 dB and SNR = 30 dB, the detection result in the range-compressed domain is shown in Fig. 5 by the proposed method. Specifically, Fig. 5(a) shows the original echo data in the range-compressed domain. Fig. 5(b) shows the moving target echo data $\hat{\mathbf{D}}$ based on the proposed NRPCA method in the range-compressed domain. The static clutter signal remaining after the moving target extraction is shown in Fig. 5(c). In Fig. 5, the moving target can effectively be separated from the clutter background by the proposed method. Moreover, the separated moving target data do not contain a residual ground clutter.

By the mention in the previous part, the factor with the greatest influence on the NRPCA and RPCA is the interchannel correlation and sparsity. As the small SNR will lead to the deterioration of the interchannel correlation and sparsity, the NRPCA and RPCA methods are compared when SNR = 10 dB.

For this purpose, the noise tolerances of the RPCA and NRPCA are shown in Fig. 6. Fig. 6(a) shows that some noise points are misjudged as the target by using the RPCA on the condition of low noise, which could lead to a high false-alarm probability. However, it can be seen that there are fewer noise points in Fig. 6(b) when using the NRPCA method. Therefore, the performance of the NRPCA is better than that of the RPCA in the case of the same noise.

Fig. 7 shows the interferometric phase of two adjacent channels for moving target 1. In Section III, we know that

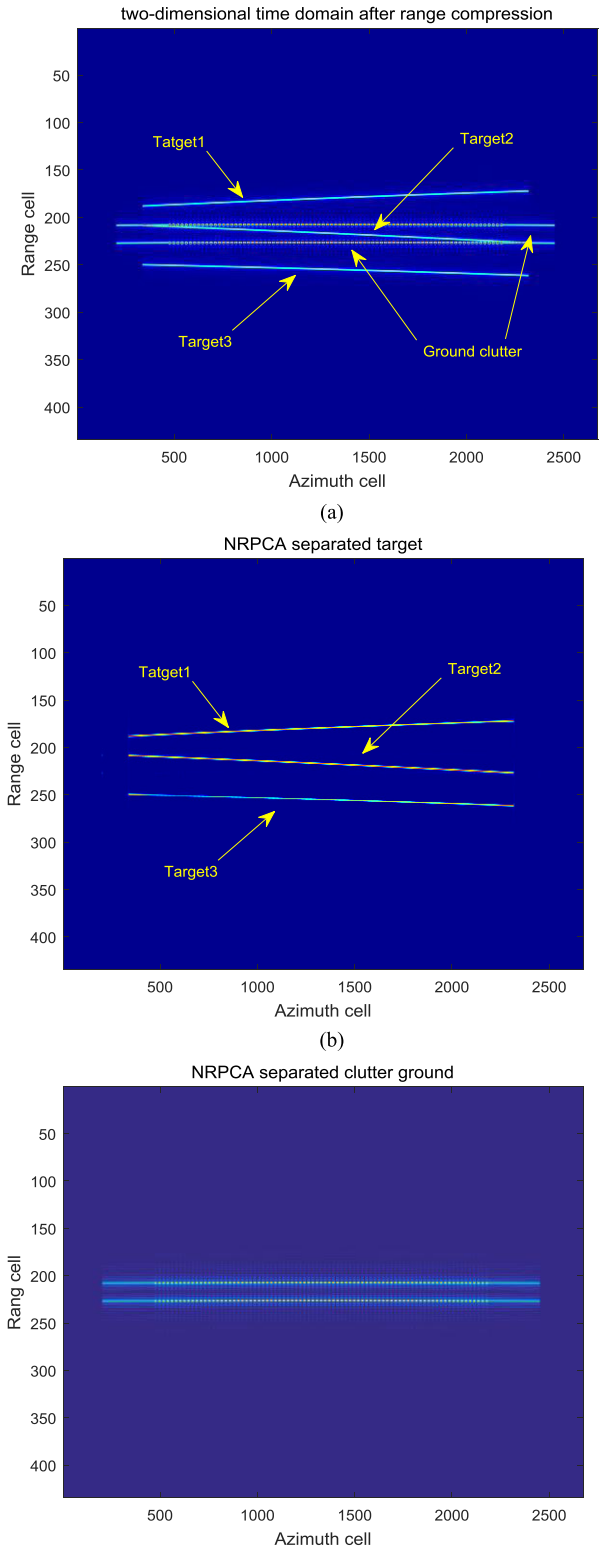


Fig. 5. Signal to be detected in the range-compressed domain based on the proposed method. (a) Echo data $\mathbf{Z}$ in the range-compressed domain. (b) Moving target echo data $\hat{\mathbf{D}}$. (c) Remaining static clutter signal $\hat{\mathbf{A}}$.

the phase information of the RPCA is destroyed such that the interferometric phase cannot be gotten by the RPCA. Hence, we take the ATI method to compare it with the NRPCA method. Under the influence of noise, the interferometric phase

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

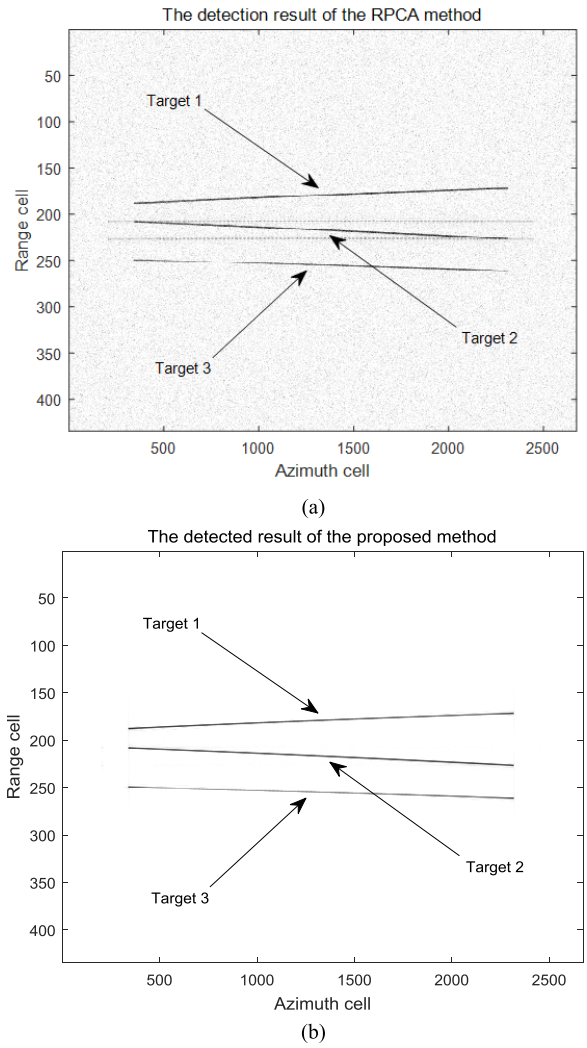


Fig. 6. Comparison of the proposed method and RPCA for SNR = 10 dB. Detection results for (a) the RPCA method and (b) the proposed method.

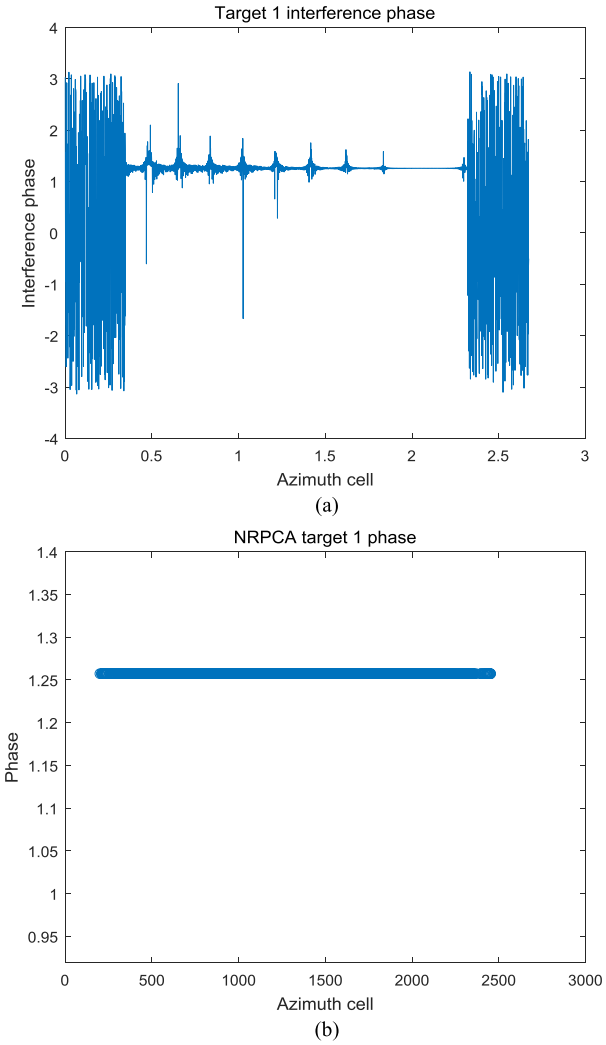


Fig. 7. Interferometric phase of two adjacent channels. (a) ATI method. (b) Proposed method.

of the ATI method exhibits fluctuations in Fig. 7(a). On the contrary, the interferometric phase of the NRPCA method is more robust under the same conditions in Fig. 7(b).

Fig. 8 shows the detection results for SCR $= -30$ dB in ideal conditions. It can be found that the position of the moving target cannot be obtained in the original data because the clutter energy is far greater than the energy of the moving target. Even so, as shown in Fig. 8(b), the moving target can still be effectively detected by the NRPCA method. Thus, combined with Fig. 5, we obtain that the NRPCA method is not sensitive for SCR. It is worth noting that the moving target detection results based on the RPCA can also be used to detect a target that is submerged in a strong clutter background. However, the results cannot reflect the original parameter characteristics of the target because its phase has been destroyed. On the contrary, the phase information can be contained in the envelope by using the proposed method, and the data of the moving target are easily recovered. Thus, the velocity of the moving target can be effectively estimated in this manner.

The velocity of the moving target can be estimated by using the different methods as shown in Fig. 9. It is generally known that the performance of the CS methods is related to the divided discrete grid points, such as the orthogonal matching pursuit (OMP) method. If the real velocity does not fall on the lattice point of the division, then there will be a deviation from the real velocity. Conversely, the velocity is estimated in the continuous domain by the proposed method and ATI method. Therefore, the grid errors do not exist for these two methods.

In Figs. 10–12, an experiment with X-band SAR raw data has been processed to show the effectiveness of the proposed method in the area of interest (AOI). The number of channels is three, and the channel spacing is 0.6 m for this radar. The platform speed is $V = 120$ m/s, and the PRF is 1000 Hz. The data in the range-Doppler domain are shown in Fig. 10(a). There are three moving targets in the clutter background. The top target with a 12.8-dB signal-to-clutter and noise ratio (SCNR) has a strong energy, whereas the surrounding clutter energy is weak. The energy of the middle target is less than that of the top target. The bottom target with SCNR $= -5.8$ dB

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

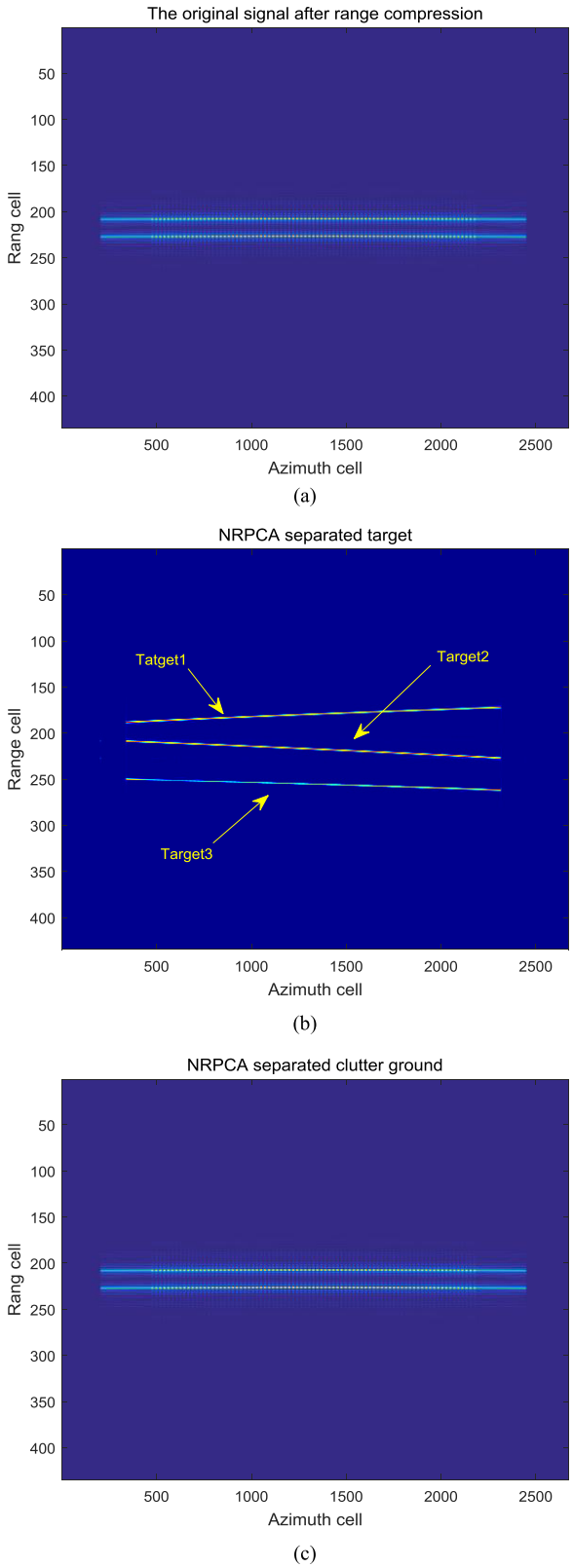


Fig. 8. Detection results for the proposed method at SCR $= -30$ dB. (a) Echo data $\mathbf{Z}$ in the range-compressed domain. (b) Moving target echo data $\hat{\mathbf{D}}$. (c) Remaining static clutter signal $\hat{\mathbf{A}}$.

has a weak energy, whereas the surrounding clutter energy is strong. Specifically, the SCNRs of all moving targets are shown in Table II.

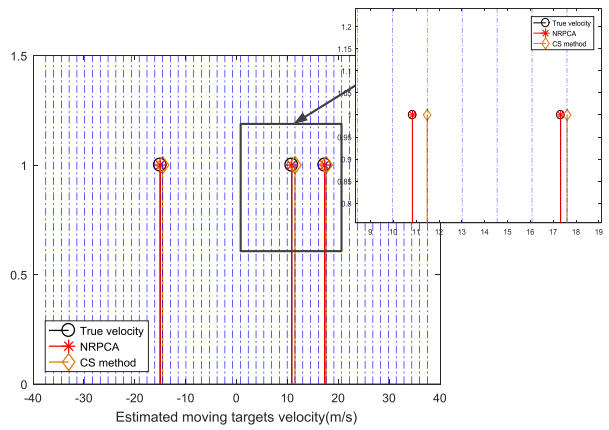


Fig. 9. Speed of the moving target as estimated by different methods.

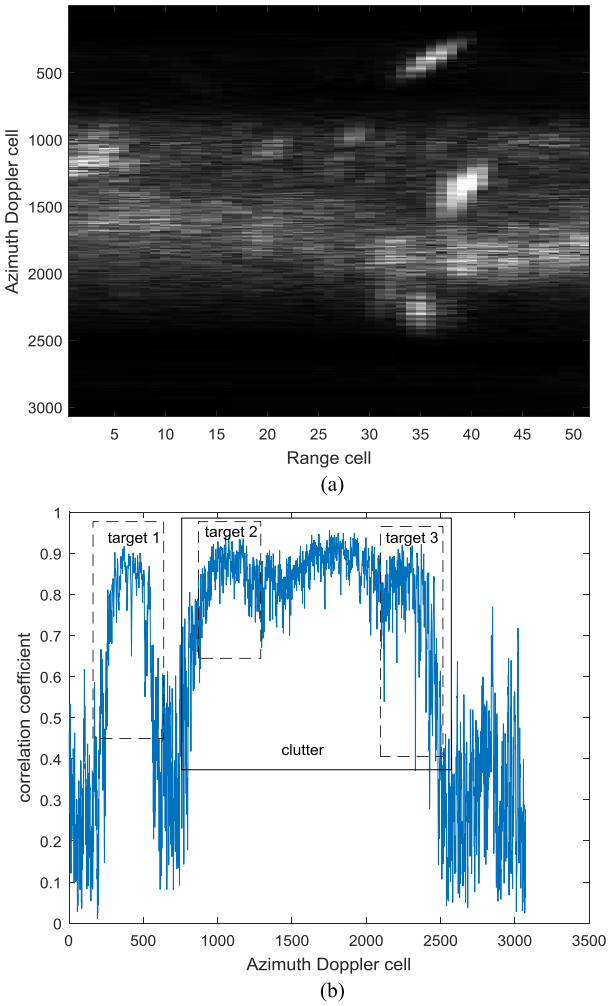


Fig. 10. Original signal and the coherence magnitude. (a) Original signal. (b) Coherence magnitude versus Doppler frequency.

In [41], the RIP condition is equal to the correlation coefficient of the measurement matrix. Therefore, a lower correlation coefficient will cause that the moving targets are extracted with a higher probability by the RPCA and NRPCA methods. Fig. 10(b) shows only a quantitative analysis of the correlation coefficient. In Fig. 10(b), the correlation curve

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

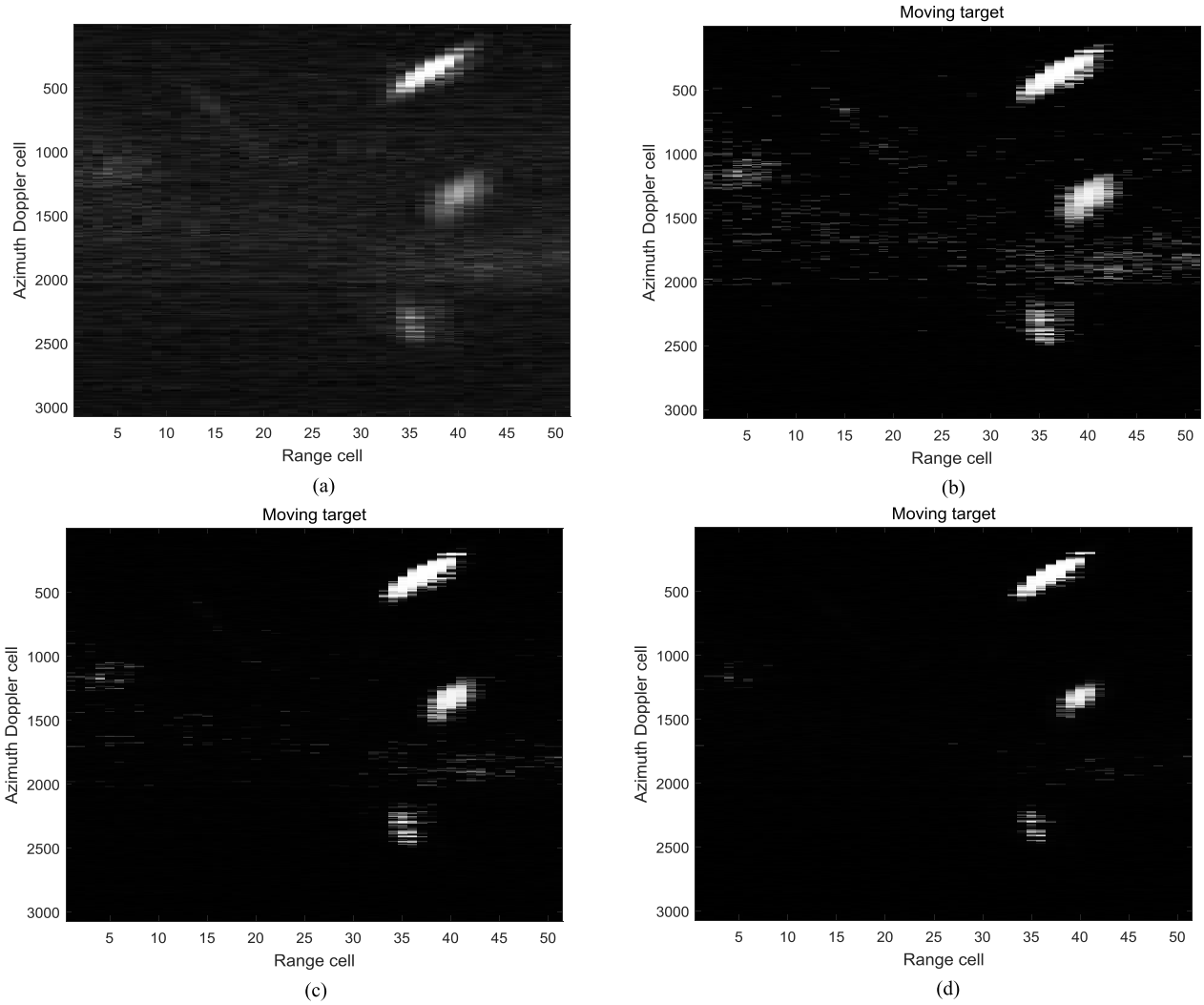


Fig. 11. Detection results for different methods. (a) Detection result using STAP. (b) Detection result using RPCA when k is 5000 points. (c) Detection result using RPCA when k is 2000 points. (d) Detection result using the proposed method.

TABLE II
SCNRs OF MOVING TARGETS

Top target	Middle target	Bottom target
12.8dB	17.3 dB	-5.8dB

exhibits a perturbation. The perturbation of the correlation curve reflects that the registration of the channel is poor, and the noise power has an impact on the correlation curve. Due to the existence of noise and channel error, the correlation of the clutter is oscillating from 0.9 to 1. Since high fluctuations in the correlation of the moving target, the correlation of the target is oscillating from 0.7 to 0.9. As the correlation of the targets is lower than that of the clutter, the moving targets can be separated from the clutter by the RPCA and NRPCA methods.

The detection results of different methods are shown in Fig. 11, where the x-axis table "Range cell" and the y-axis table "Azimuth cell" denote the grid point in the range time domain and the grid point in the azimuth time domain, respectively. "Azimuth Doppler cell" represents the grid point in the azimuth Doppler domain.

By using the traditional STAP method, the results are shown in Fig. 11(a). The detection results for the RPCA method are shown in Fig. 11(b), when the value of the sparsity order k is 5000. The detection results based on the RPCA method are shown in Fig. 11(c) when the value of the sparsity order k is 2000. The NRPCA method is shown in Fig. 11(d).

It is well known that the performance of STAP depends on the selection of the sample. Compared with STAP, the RPCA method is more effective when the channel correlation is better and the energy of the moving target signal is strong. In Fig. 11(b) and (c), there are fewer false points, but the moving targets are missing when the value of the sparsity order k is lower. When the value of the sparsity order k is greater, the moving target is closer to the real target, but there are more false points. It means that the selection of the sparsity order k will impact the performance of the RPCA method. On the contrary, it seems that the proposed method is less affected by noise and does not require to select the value of the sparsity order k, as shown in Fig. 11(d).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

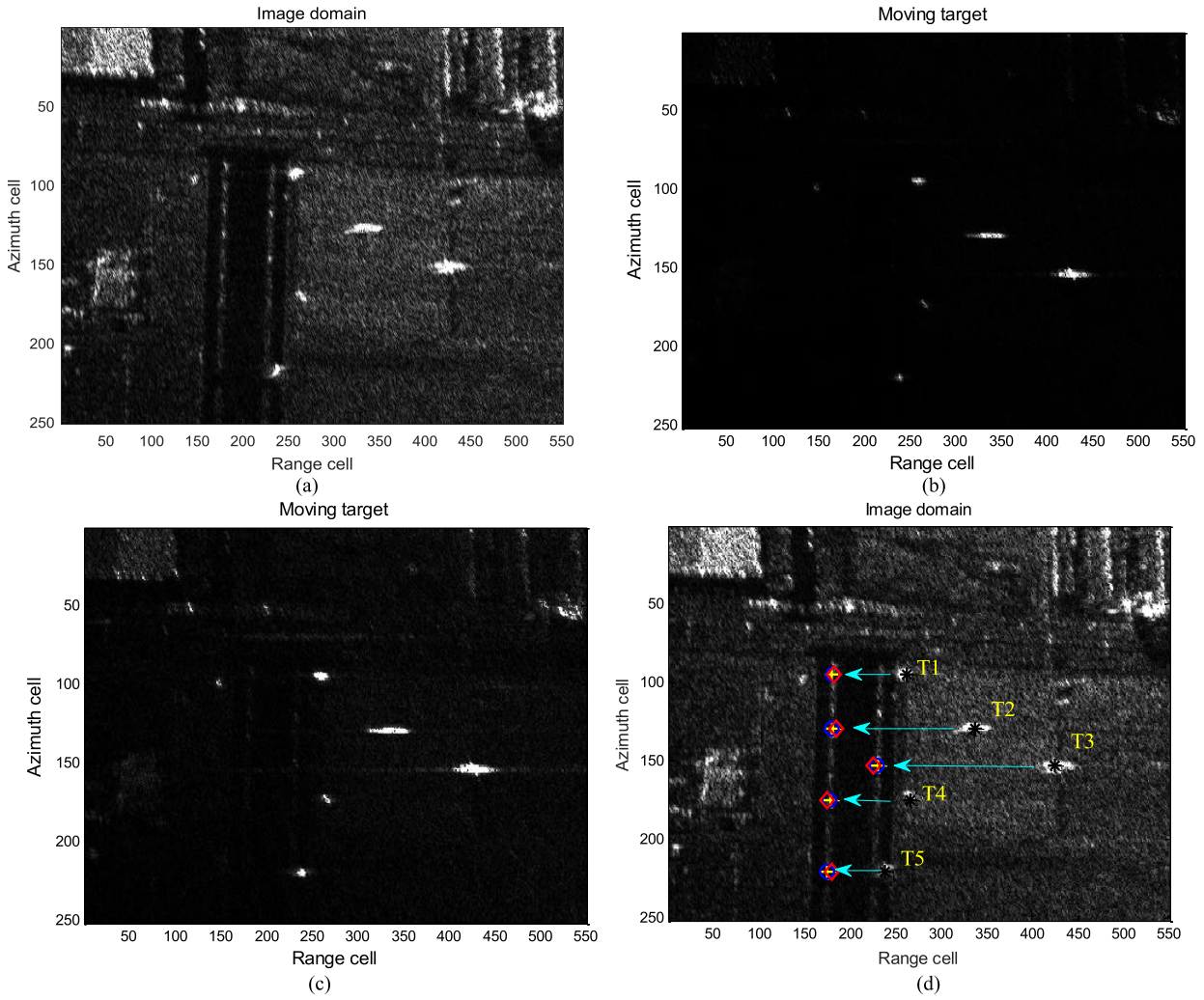


Fig. 12. Detection results in the image domain. (a) Original signal. (b) Detection result for the proposed method in the image domain. (c) Detection result for the STAP method in the image domain. (d) Parameter estimation and relocation result.

To this end, all three methods yield good results for the above two targets. Unfortunately, the residue of the strong clutter energy around the moving target at the bottom caused by the channel mismatch will be misjudged as a moving target. Thus, the RPCA may cause the false points. However, after the zero value of the speed is removed by (30) and the echo data for the moving targets are restructured by (9), the NRPCA method can reduce the false-alarm probability since targets with zero velocity are removed.

In Fig. 12, the SAR raw data of the first channel have been processed to evaluate the detection results of the proposed method in the image domain. There are five controlled vehicles in the ways (the real radial velocities of the moving targets are -0.6119 m/s ($T1$), -1.5938 m/s ($T2$), -1.8598 m/s ($T3$), -0.7173 m/s ($T4$), and -0.2829 m/s ($T5$), respectively). The AOI data in the image domain are shown in Fig. 12(a), and the detection results for the proposed method and the STAP method are shown in Fig. 12(b) and (c), respectively. It can be observed that the clutter suppression performance of the proposed method is better than that of the STAP method because the discrete clutter could not be suppressed by the STAP method. The radial velocities of the moving targets estimated by different methods are shown in Table III.

TABLE III
ESTIMATED RADIAL VELOCITIES OF THE MOVING TARGETS

	T1(m/s)	T2(m/s)	T3(m/s)	T4(m/s)	T5(m/s)
Real radial velocity	−0.6119	−1.5938	−1.8598	−0.7173	−0.2829
NRPCA	-0.6120	-1.5939	-1.8586	-0.7174	-0.2823
CS	-0.6111	-1.6011	-1.8712	-0.7236	-0.2736
ATI	-0.6124	-1.5942	-1.8583	-0.7170	-0.2826

Then, the repositioning of the moving target after velocity estimation based on the NRPCA, CS, and ATI methods is plotted in yellow plus sign points, red diamond points, and blue circle points of Fig. 12(d), respectively. The above figure seems that the proposed method and the ATI method have similar relocated results. However, the result using the CS method is different from the proposed method and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

ATI method. The results shown in Fig. 12 are consistent with the above conclusion that the CS method has grid errors. To this end, the proposed method is superior to the traditional methods.

VI. Conclusion

An NRPCA for SAR systems is proposed in this article. The proposed method has released the nonsparse limitations of the moving target echo and avoided the destroyed phase information in the existing algorithm. Not only can this approach effectively separate the target from the background clutter but it can also accurately estimate the radial velocity of the moving target. In addition, the selection of the sparsity order k is avoided, and the proposed method is not sensitive to SCR. A simulation and analysis illustrate the effectiveness of the proposed method.

Appendix A

Proof: Let us denote the objective function of a minimization problem as $f(x)$. When $\hat{x}$ is a solution, this minimization problem is equivalent to

$$f\left(\hat{x}+\alpha\left(x-\hat{x}\right)\right)\geq f\left(\hat{x}\right)\quad \forall\alpha\in(0,1),\ x\in\mathbb{R}^N. \tag{33}$$

By substituting $f(x)$, which is the objective function in problem (25), into (33) and using the convexity of the atomic norm, it follows that

$$\begin{aligned}\lambda\cdot\|\mathbf{D}\|_{\mathcal{A}}+\frac{\mu}{2}\cdot\|\mathbf{A}_{k+1}-(\mathbf{Z}-\mathbf{D}+\mathbf{Y}_k/\mu)\|_F^2\\ \geq\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}+\frac{\mu}{2}\cdot\|\mathbf{A}_{k+1}-(\mathbf{Z}-\hat{\mathbf{D}}+\mathbf{Y}_k/\mu)\|_F^2\end{aligned} \tag{34}$$

namely,

$$\begin{aligned}\lambda\cdot\|\mathbf{D}\|_{\mathcal{A}}-\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}\geq\mu\langle\mathbf{A}_{k+1}-\mathbf{Z}-\mathbf{Y}_k/\mu+\hat{\mathbf{D}}/2,\hat{\mathbf{D}}\rangle\\ -\mu\langle\mathbf{A}_{k+1}-\mathbf{Z}-\mathbf{Y}_k/\mu+\mathbf{D}/2,\mathbf{D}\rangle.\end{aligned} \tag{35}$$

Equation (35) is necessary and sufficient when $\hat{\mathbf{D}}$ is the solution. Note that (35) can be rewritten as follows:

$$\begin{aligned}\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}-\mu\langle\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\hat{\mathbf{D}},\hat{\mathbf{D}}\rangle\\ \leq\inf_{\mathbf{D}}\{\lambda\cdot\|\mathbf{D}\|_{\mathcal{A}}-\mu\langle\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\mathbf{D},\mathbf{D}\rangle\}.\end{aligned} \tag{36}$$

From the definition of the dual norm of the atomic norm, it is yielded that

$$\begin{aligned}\inf_{\mathbf{D}}\{\lambda\cdot\|\mathbf{D}\|_{\mathcal{A}}-\mu\langle\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\mathbf{D},\mathbf{D}\rangle\}\\ =\begin{cases}0, & \|\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\mathbf{D}\|_{\mathcal{A}}\leq\lambda/\mu\\ -\infty, & \text{otherwise.}\end{cases}\end{aligned} \tag{37}$$

Therefore, the first condition for the solution is $\|\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\mathbf{D}\|_{\mathcal{A}}\leq\lambda/\mu$. Then, it follows from (36) that

$$\begin{aligned}\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}-\mu\langle\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\hat{\mathbf{D}},\hat{\mathbf{D}}\rangle\\ =\inf_{\mathbf{D}}\{\lambda\cdot\|\mathbf{D}\|_{\mathcal{A}}-\mu\langle\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\mathbf{D},\mathbf{D}\rangle\}=0\end{aligned} \tag{38}$$

namely,

$$\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}=\mu\langle\mathbf{Z}+\mathbf{Y}_k/\mu-\mathbf{A}_{k+1}-\hat{\mathbf{D}},\hat{\mathbf{D}}\rangle. \tag{39}$$

By using (29), (39) can be rewritten as

$$\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}=\langle\mathbf{Y}',\hat{\mathbf{D}}\rangle. \tag{40}$$

Suppose that $\hat{\mathbf{D}}=\sum_{\mathbf{l}_a\in\mathbf{S}}\sigma_a\mathbf{l}_a(\upsilon)\mathbf{r}_a^T+\sum_{\mathbf{l}_b\notin\mathbf{S}}\sigma_b\mathbf{l}_b(\upsilon)\mathbf{r}_b^T$. One can obtain

$$\begin{aligned}\langle\mathbf{Y}',\hat{\mathbf{D}}\rangle&=\sum_{\mathbf{l}_a\in\mathbf{S}}\sigma_a\left\langle\mathbf{l}_a(\upsilon)\mathbf{r}_a^T,\mathbf{Y}'\right\rangle+\sum_{\mathbf{l}_b\notin\mathbf{S}}\sigma_b\left\langle\mathbf{l}_b(\upsilon)\mathbf{r}_b^T,\mathbf{Y}'\right\rangle\\ &=\lambda\sum_{\mathbf{l}_a\in\mathbf{S}}\sigma_a\mathbf{r}_a+\sum_{\mathbf{l}_b\notin\mathbf{S}}\sigma_b\left\langle\mathbf{l}_b(\upsilon)\mathbf{r}_a^T,\mathbf{Y}'\right\rangle.\end{aligned} \tag{41}$$

The left-hand side of (40) can be expressed as

$$\lambda\cdot\|\hat{\mathbf{D}}\|_{\mathcal{A}}=\lambda\sum_{\mathbf{l}_a\in\mathbf{S}}\sigma_a\mathbf{r}_a+\lambda\sum_{\mathbf{l}_b\notin\mathbf{S}}\sigma_b\mathbf{r}_b. \tag{42}$$

If $\exists\mathbf{l}_b\notin\mathbf{S}$ such that $\sigma_b\neq0$, then there is a contradiction that the last two equations cannot hold since $\sum_{\mathbf{l}_b\notin\mathbf{S}}\sigma_b\langle\mathbf{l}_b(\upsilon)\mathbf{r}_b^T,\mathbf{Y}'\rangle<\lambda\sum_{\mathbf{l}_b\notin\mathbf{S}}\sigma_b\mathbf{r}_b$. Thus, $\forall\mathbf{l}_b\notin S,\sigma_b=0$, which means that $\mathbf{S}$ is the support set of $\hat{\mathbf{D}}$.

Appendix B

The convergence problem of the ADMM is shown in [42]. The algorithm solves the problems of the form

$$\begin{aligned}&\min_{\mathbf{x},\mathbf{z}}\ f(\mathbf{x})+g(\mathbf{z})\\ &\text{s.t. }\mathbf{A}\mathbf{x}+\mathbf{B}\mathbf{z}=\mathbf{c}\end{aligned} \tag{43}$$

where $\mathbf{x}\in\mathbb{R}^n$, $\mathbf{z}\in\mathbb{R}^m$, $\mathbf{A}\in\mathbb{R}^{p\times n}$, $\mathbf{B}\in\mathbb{R}^{p\times m}$, and $\mathbf{c}\in\mathbb{R}^p$.

Assumption 1: The functions $f\colon\mathbf{R}^n\to\mathbf{R}\cup\{+\infty\}$ and $g\colon\mathbf{R}^m\to\mathbf{R}\cup\{+\infty\}$ are closed, proper, and convex.

Assumption 2: The augmented Lagrange L_0 has a saddle point.

Under *Assumptions 1* and *2*, the ADMM iterations satisfy the following conclusions.

1) *Residual Convergence:* $\mathbf{r}^k\to0$ as $k\to\infty$ where defining the residual $\mathbf{r}^k=\mathbf{A}\mathbf{x}^k+\mathbf{B}\mathbf{z}^k-\mathbf{c}$, i.e., the iteration approaches feasibility.
2) *Objective Convergence:* $f(x^k)+g(z^k)\to p^*$ as $k\to\infty$, i.e., the objective function of the iterations approaches the optimal value.
3) *Dual Variable Convergence:* $y^k\to y^*$ as $k\to\infty$, where y^* is a dual optimal point.

If two assumptions are satisfied in the proposed optimization problem, the phase of $\mathbf{D}$ is convergent. Explicitly, $f(\mathbf{D})$ and $g(\mathbf{A})$ are closed, proper, and convex in (19) and (25) in this article. Therefore, assumption 1 is satisfied. $\mathbf{Y}$ is the maximum value in (28). $\mathbf{A}$ and $\mathbf{D}$ are the minimum values in (19) and (25), respectively. Therefore, the augmented Lagrange $L_0(\mathbf{A},\mathbf{D},\mathbf{Y})$ has a saddle point.

In conclusion, the phase of $\mathbf{D}$ is convergent.

References

[1] X. Zhang, G. Liao, S. Zhu, C. Zeng, and Y. Shu, "Geometry-information-aided efficient radial velocity estimation for moving target imaging and location based on radon transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1105–1117, Feb. 2015.

[2] X. Zhang, G. Liao, S. Zhu, Y. Gao, and J. Xu, "Geometry-information-aided efficient motion parameter estimation for moving-target imaging and location," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 155–159, Jan. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.

[3] A. Budillon, V. Pascazio, and G. Schirinzi, "Estimation of radial velocity of moving targets by along-track interferometric SAR systems," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 3, pp. 349–353, Jul. 2008.

[4] V. Pascazio, G. Schirinzi, and A. Farina, "Moving target detection by along-track interferometry," in *Proc. Scanning Present Resolving Future, IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sydney, NSW, Australia, vol. 3027, Jul. 2001, pp. 3024–3026.

[5] C. H. Gierull, "Azimuth positioning of moving targets in two-channel SAR by direction-of-arrival estimation," *Electron. Lett.*, vol. 40, no. 21, pp. 1380–1381, Oct. 2004.

[6] Y. Huang, G. Liao, J. Li, and J. Xu, "Ground moving target indication and parameter estimation with single channel for SAR system," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Beijing, China, Jul. 2016, pp. 6417–6420.

[7] G. Li, X.-G. Xia, J. Xu, and Y.-N. Peng, "A velocity estimation algorithm of moving targets using single antenna SAR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 45, no. 3, pp. 1052–1062, Jul. 2009.

[8] F. Zhou, R. Wu, M. Xing, and Z. Bao, "Approach for single channel SAR ground moving target imaging and motion parameter estimation," *IET Radar, Sonar Navigat.*, vol. 1, no. 1, pp. 59–66, 2007.

[9] B. Liu, K. Yin, Y. Li, F. Shen, and Z. Bao, "An improvement in multichannel SAR-GMTI detection in heterogeneous environments," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 810–827, Feb. 2015.

[10] B. Liu, T. Wang, Y. Li, F. Shen, and Z. Bao, "Effects of Doppler aliasing on baseline estimation in multichannel SAR-GMTI and solutions to address these effects," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6471–6487, Oct. 2014.

[11] C. H. Gierull and I. C. Sikaneta, "Raw data based two-aperture SAR ground moving target indication," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Toulouse, France, Jul. 2003, pp. 1032–1034.

[12] E. Chapin and C. W. Chen, "Airborne along-track interferometry for GMTI," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 24, no. 5, pp. 13–18, May 2009.

[13] D. Cerutti-Maori and I. Sikaneta, "A generalization of DPCA processing for multichannel SAR/GMTI radars," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 560–572, Jan. 2013.

[14] A. Budillon, V. Pascazio, and G. Schirinzi, "Multichannel along-track interferometric SAR systems: Moving targets detection and velocity estimation," *Int. J. Navigat. Observ.*, vol. 2008, Feb. 2008, Art. no. 310656.

[15] J. H. G. Ender, "Space-time processing for multichannel synthetic aperture radar," *Electron. Commun. Eng. J.*, vol. 11, no. 1, pp. 29–38, Feb. 1999.

[16] X. Li, M. Xing, X.-G. Xia, G.-C. Sun, Y. Liang, and Z. Bao, "Deramp space-time adaptive processing for multichannel SAR systems," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 8, pp. 1448–1452, Aug. 2014.

[17] I. Stojanovic and W. C. Karl, "Imaging of moving targets with multistatic SAR using an overcomplete dictionary," *IEEE J. Sel. Topics Signal Process.*, vol. 4, no. 1, pp. 164–176, Feb. 2010.

[18] B. Recht, M. Fazel, and P. A. Parrilo, "Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization," *SIAM Rev.*, vol. 52, no. 3, pp. 471–501, Jan. 2010.

[19] Y. Peng, J. Suo, Q. Dai, and W. Xu, "Reweighted low-rank matrix recovery and its application in image restoration," *IEEE Trans. Cybern.*, vol. 44, no. 12, pp. 2418–2430, Dec. 2014.

[20] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1–37, May 2011.

[21] T. Bouwmans, S. Javed, H. Zhang, Z. Lin, and R. Otazo, "On the applications of robust PCA in image and video processing," *Proc. IEEE*, vol. 106, no. 8, pp. 1427–1457, Aug. 2018.

[22] N. Vaswani, T. Bouwmans, S. Javed, and P. Narayanamurthy, "Robust subspace learning: Robust PCA, robust subspace tracking, and robust subspace recovery," *IEEE Signal Process. Mag.*, vol. 35, no. 4, pp. 32–55, Jul. 2018.

[23] F. Biondi, "Low rank plus sparse decomposition of synthetic aperture radar data for maritime surveillance," in *Proc. 4th Int. Workshop Compressed Sens. Theory Appl. Radar, Sonar Remote Sens. (CoSeRa)*, Aachen, Germany, Sep. 2016, pp. 75–79.

[24] L. Borcea, T. Callaghan, and G. Papanicolaou, "Synthetic aperture radar imaging and motion estimation via robust principal component analysis," *SIAM J. Imag. Sci.*, vol. 6, no. 3, pp. 1445–1476, Jan. 2013.

[25] M. Yasin, M. Cetin, and A. S. Khwaja, "SAR imaging of moving targets by subaperture based low-rank and sparse decomposition," in *Proc. 25th Signal Process. Commun. Appl. Conf. (SIU)*, May 2017, pp. 1–4.

[26] Y. Huang and G. Liao, "MIMO SAR GMTI with RPCA based method," in *Proc. IEEE 5th Asia–Pacific Conf. Synth. Aperture Radar (APSAR)*, Singapore, Sep. 2015, pp. 827–830.

[27] Y. Huang, G. Liao, J. Xu, and D. Yang, "MIMO SAR OFDM chirp waveform design and GMTI with RPCA based method," *Digit. Signal Process.*, vol. 51, pp. 184–195, Apr. 2016.

[28] F. Biondi, "A polarimetric extension of low-rank plus sparse decomposition and radon transform for ship wake detection in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 75–79, Jan. 2018.

[29] Q. Li, H. Yan, L. Wu, and R. Wang, "Robust PCA for ground moving target indication in wide-area surveillance radar system," *J. Oper. Res. Soc. China*, vol. 1, no. 1, pp. 135–153, Mar. 2013.

[30] D. Yang, G. Liao, S. Zhu, and X. Yang, "RPCA based moving target detection in strong clutter background," in *Proc. IEEE Radar Conf. (RadarCon)*, Arlington, TX, USA, May 2015, pp. 1487–1490.

[31] D. Yang, X. Yang, G. Liao, and S. Zhu, "Strong clutter suppression via RPCA in multichannel SAR/GMTI system," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2237–2241, Nov. 2015.

[32] J. Li, Y. Huang, G. Liao, and J. Xu, "Moving target detection via efficient ATI-GoDec approach for multichannel SAR system," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 9, pp. 1320–1324, Sep. 2016.

[33] A. H. Oveis and M. A. Sebt, "Dictionary-based principal component analysis for ground moving target indication by synthetic aperture radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1594–1598, Jul. 2017.

[34] Z. Liu and L. Vandenberghe, "Interior-point method for nuclear norm approximation with application to system identification," *SIAM J. Matrix Anal. Appl.*, vol. 31, no. 3, pp. 1235–1256, Jan. 2009.

[35] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Jan. 2010.

[36] E. J. Candès and C. Fernandez-Granda, "Towards a mathematical theory of super-resolution," *Commun. Pure Appl. Math.*, vol. 67, no. 6, pp. 906–956, Jun. 2014.

[37] B. S. He, H. Yang, and S. L. Wang, "Alternating direction method with self-adaptive penalty parameters for monotone variational inequalities," *J. Optim. Theory Appl.*, vol. 106, no. 2, pp. 337–356, Aug. 2000.

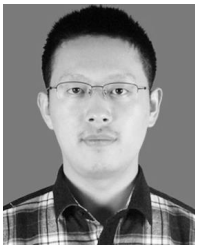
[38] S. L. Wang and L. Z. Liao, "Decomposition method with a variable parameter for a class of monotone variational inequality problems," *J. Optim. Theory Appl.*, vol. 109, no. 2, pp. 415–429, May 2001.

[39] G. Tang, B. N. Bhaskar, P. Shah, and B. Recht, "Compressed sensing off the grid," *IEEE Trans. Inf. Theory*, vol. 59, no. 11, pp. 7465–7490, Nov. 2013.

[40] C. Michael and B. Stephen. (Mar. 2014). *CVX: MATLAB Software for Disciplined Convex Programming, Version 2.0*. [Online]. Available: http://cvxr.com/cvx/citing/

[41] D. L. Donoho, M. Elad, and V. N. Temlyakov, "Stable recovery of sparse overcomplete representations in the presence of noise," *IEEE Trans. Inf. Theory*, vol. 52, no. 1, pp. 6–18, Jan. 2006.

[42] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.



Yifan Guo was born in Henan, China, in 1990. He received the B.S. degree in information engineering from the North China University of Water Resources and Electric Power, Zhengzhou, China, in 2012, and the M.S. degree in electrical engineering from Xidian University, Xi'an, China, in 2015, where he is pursuing the Ph.D. degree with the National Laboratory of Radar Signal Processing.

His research interests include bistatic multiple-input multiple-output (Bi-MIMO) radar, synthetic aperture radar (SAR), ground moving target indication (GMTI), and space-time adaptive processing (STAP).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.



Guisheng Liao (Senior Member, IEEE) received the B.S. degree from Guangxi University, Nanning, Guangxi, China, in 1985, and the M.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 1990 and 1992, respectively.

He is a Professor with the National Key Laboratory of Radar Signal Processing, Xidian University, where he is also the Dean of the School of Electronic Engineering. His research interests include space-time adaptive processing, array signal processing, synthetic aperture radar (SAR) ground moving target indication, and distributed small satellite SAR system design.

Dr. Liao is a member of the National Outstanding Person and the Cheung Kong Scholars in China.



Jun Li (Member, IEEE) received the B.S. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 1994, the M.S. degree from the Guilin University of Electronic Technology, Guilin, China, in 2002, and the Ph.D. degree in information and communication engineering from Xidian University, Xi'an, China, in 2005.

After receiving his Ph.D. degree, he joined Xidian University, where he is a Professor with the National Laboratory of Radar Signal Processing. From November 2009 to October 2010, he was a Visiting Scholar with the Department of Electronic and Electrical Engineering, University College London, London, U.K. His research interests include array signal processing, space-time adaptive signal processing, multiple-input multiple-output (MIMO) radar, and quantum sensors.

Dr. Li received the 2014 IET Innovation Award, the Highly Commended Award, and the Excellent Paper Awards from Radar 2011.



Xixi Chen was born in Hunan, China, in 1989. She received the B.S. degree in information engineering from Xiangtan University, Xiangtan, China, in 2012, and the M.S. degree in electrical engineering from Xidian University, Xi'an, China, in 2015. She is pursuing the Ph.D. degree in electronic science and engineering with the National University of Defense Technology, Changsha, China.

Her research interests include information geometry and statistical signal processing, weak target detection, and waveform design.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:24:10 UTC from IEEE Xplore. Restrictions apply.